

Arbitrage Exists, but Not at Retail: Measuring the Retail Accessibility of Cryptocurrency Arbitrage

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Abstract

We measure whether cryptocurrency arbitrage is accessible to a retail participant using commodity infrastructure, and report primarily negative results. Streaming Binance best-bid/offer data for 2.74 days (66,476,148 ticks) across five connected spot markets, an event-driven negative-cycle detector found zero executable triangular-arbitrage cycles once an execution-realistic simultaneity constraint (0.2-second maximum quote-age spread) was imposed. A detector without this constraint over-reports phantom opportunities—here up to +51.6 basis points—assembled from non-simultaneous quotes, reconciling prior work that finds arbitrage captured by co-located bots with prior work that finds it unexploitable at retail. On the funding-carry track, a two-year backtest over 151 perpetual-futures symbols reproduces a pitfall: a “no opportunity” conclusion under a curated major-coin universe reverses to positive validation-window profit under a volume-ranked broad universe, so the conclusion is driven by universe selection. A post-hoc placebo test ($N = 1000$, seed = 42) then finds the funding entry rule’s profit not distinguishable from random entry timing (37.8th percentile; one-sided $p = 0.622$): in attribution terms, the return is beta to a carry factor, not alpha from timing skill. In our sample, arbitrage-like returns exist but are not accessible to retail as a skill-based edge.

JEL classification: G12 (Asset Pricing); G14 (Information and Market Efficiency); G23 (Non-bank Financial Institutions)

Keywords: cryptocurrency arbitrage; triangular arbitrage; funding-rate carry; limits to arbitrage; market-timing skill

1. Introduction

Two recent empirical findings on Binance appear to be in tension. Grimberg et al. (2020) document algorithmic bots capturing triangular arbitrage at roughly +14.4 basis points on ~2.71% of trades, while Muck et al. (2025) find that triangular arbitrage, though present in the order book, is difficult to exploit once transaction costs and execution risk are accounted for. The two are reconciled by a single variable that is usually left implicit: quote simultaneity. Whoever can act on three quotes as if they are contemporaneous captures the edge; whoever cannot is measuring a mirage. This paper makes that variable explicit and asks what a retail participant—commodity hardware, public WebSocket feeds, taker fees—can actually reach.

Our contributions are four:

- (i) A large real-time retail measurement of triangular-arbitrage accessibility. We stream Binance best-bid/offer data continuously for 2.74 days (66,476,148 ticks) and report zero executable detections under a 0.2-second simultaneity constraint.
- (ii) A methodological finding on phantom arbitrage. We show, both by construction and by controlled experiment, that a detector computing cycle returns without a simultaneity constraint over-reports opportunities (here +51.6 bp) that vanish once quotes must be contemporaneous—clarifying why naive measurements overstate retail arbitrage.
- (iii) A reproduction of universe-selection dependence in funding-carry backtests. The same rule yields zero validation-window entries on a curated major-coin universe but positive profit on a volume-ranked broad universe—the conclusion is a function of universe choice, not of a market fact.
- (iv) A placebo test separating carry exposure from timing skill. Holding universe, period, per-symbol entry count, and holding rule fixed while randomizing entry timing, the actual signal does not beat random entry ($p = 0.622$).

Throughout we follow a discipline of reporting negative and reversed results rather than suppressing them. The triangular result is stated as an observation over a finite window, not as a universal claim of non-existence; the funding result is stated as an absence of evidence for timing skill in this sample, not as a proof that skill is impossible.

2. Data

Triangular track (primary evidence). We consume Binance spot `bookTicker` streams (`wss://stream.binance.com`) for a connected set of five markets—BTC/USDT, ETH/USDT, SOL/USDT, ETH/BTC, SOL/ETH—chosen because they form closable triangles; the cross pairs ETH/BTC and SOL/ETH exist only on the spot venue. A runner streamed continuously from 2026-07-16 02:55:34 KST (UTC+9) and was snapshotted read-only at 2026-07-18 11:39:12 UTC,

a span of about 2.74 days, having processed 66,476,148 ticks. The snapshot is the sole source for triangular figures.

Funding-carry track. We use `fetch_funding_rate_history` from two exchanges via CCXT, covering 2024-07-16 to 2026-07-16 (729.9 days) for 151 symbols (100 Binance USDT perpetuals, 51 Bybit). The universe is the current (2026-07) volume-ranked top-100 per exchange, replayed backward—a survivorship choice we flag in Limitations.

FX / CIP track (design verification only). An FX-spot and covered-interest-parity monitor was built and exercised on a 12-event replay fixture. Because it was never run on a live stream, we treat it strictly as engine/design verification and make no empirical claim from it.

3. Methodology

Detection. Prices are placed on spacetime nodes (`asset`, `settlement`); an arbitrage is a negative-weight cycle where each edge weight is $-\log(\text{rate})$ on the executable (fee-adjusted bid/ask) side. Detection is event-driven: on each quote update we search only cycles through the updated edge. A cycle is executable only if it passes a staleness filter—its edges’ quote-age spread must not exceed `max_ts_spread = 0.2 s`—which encodes a retail-realistic requirement that the combined quotes be near-contemporaneous. We separately record a `net_bp` for the best reconstructed cycle regardless of the staleness filter, to quantify near-misses.

Funding backtest. We replay funding history on an 8-hour grid using the same entry and exit rules the live system uses (imported, not re-implemented): enter when the break-even horizon (round-trip fee / periodic funding) is below a threshold and the trailing 30-day positive-funding ratio exceeds 0.8; exit on two consecutive negative funding prints. Entry decisions at time t use only data with timestamp $< t$ (no look-ahead). We evaluate with a walk-forward split (first year select, second year validate) and, for risk, over the full two years.

Verification design (this paper). All figures were regenerated from frozen caches at code commit 87d6c6b and stored under `paper/data/`; the walk-forward reversal was run twice and was bitwise identical (deterministic). The placebo of Section 4.4 is a post-hoc robustness check, not a pre-registered test; its simulator was validated to reproduce the production backtest engine exactly before use.

4. Results

4.1 No executable triangular arbitrage at retail

Over 2.74 days and 66,476,148 ticks, the detector ran 66,476,148 per-edge scans and returned zero executable cycles.¹ We state this as an observation over a finite window rather than a universal

¹The activity counter increments once per directed edge processed, and one event-driven cycle-detection scan runs per edge. Each Binance `bookTicker` message yields two directed edges—a sell edge (base→quote) and a buy

non-existence claim. It is consistent with Muck et al. (2025) at retail resolution and with Shleifer and Vishny (1997): the edge, where it exists, accrues to participants with capital and speed that our retail configuration lacks.

4.2 Phantom arbitrage: the cost of ignoring simultaneity (methodological)

The best near-miss recorded in the window was +51.6 bp. Because our `net_bp` is computed on the reconstructed cycle independently of the staleness filter, and because the detector’s minimum-gross filter is zero, the only filter that can reject a positive cycle is staleness. It follows logically—and we confirm by controlled experiment—that any positive near-miss recorded alongside zero executable detections must arise from a quote-age spread above 0.2 s, i.e. from non-simultaneous quotes. In the experiment, a profitable triangle whose third quote is delayed by 0.5 s yields a reconstructed cycle of +368.83 bp with a staleness spread of 0.500 s and no executable detection; the identical triangle with contemporaneous quotes is detected. The methodological implication is that a naive measurement without a simultaneity constraint over-reports retail arbitrage that does not survive execution; this is, we argue, the mechanism behind the Grimberg–Muck tension.

4.3 Funding carry: a conclusion that flips with universe selection

Applying the fixed rule with a walk-forward split, a curated major-coin universe (the 20 largest majors, BTC...FIL) produces zero validation-window entries and \$0 profit; across all threshold/exit combinations the validation profit is identically zero (0% positive). The volume-ranked broad universe (151 symbols) instead produces 12 validation entries and +\$47.44 validation profit for the selected combination, with the validation profit ranging [−\$0.60, +\$125.35] across combinations and positive in 65% of them. The reversal is deterministic across two runs. The lesson is that the earlier “no opportunity” reading was an artifact of universe selection (majors-only), not of universe size: funding opportunities concentrate in higher-funding alt-coins that a major-coin list excludes.

Table 1: Funding-carry backtest: the same rule under different universes and horizons. Source: Author’s calculations.

Universe / view	Val. entries	Annual PnL	Note
Curated major-coin 20, validation	0	\$0.00	\$0 across all combos
Volume-ranked broad 151, validation	12	+\$47.44	65% of combos positive
\$700 / \$200 per symbol, validation	—	+\$196.94	28.1% of capital
\$700 / \$200 per symbol, full two years	—	+\$27.55	regime-concentrated
\$700 / \$200, top-3 positions removed	—	+\$14.25	52% of baseline

edge (quote→base)—so the 66,476,148 figure counts edges/scans and corresponds to roughly 33.2 million order-book messages.

Under an unconstrained-capital validation-window view, the strategy looks attractive—e.g. +\$196.94/yr on \$700 deployed at \$200/symbol (28.1% of capital) —but the return is regime-concentrated: over the full two years the same \$700/\$200 configuration returns only +\$27.55/yr. Profit is also concentration-dependent: removing the top three positions cuts the full-period annual return to 52% of baseline. Drawdown is small (max -\$0.60, 0.0% at \$700/\$200; -\$2.07, 2.5% at \$2000/\$500) ; the operative risk is opportunity scarcity (longest dry spell 142.2 days, longest underwater 8.1 weeks) rather than capital loss, and no calendar month is negative. In a separate scaling stress measured at a 10-concurrent, \$200/symbol configuration over the validation window, the strategy is also slippage-tolerant, remaining positive at +20 bp of added round-trip slippage (+\$234.35/yr, versus +\$265.82/yr at 0 bp).

4.4 The funding signal shows no timing skill beyond exposure (central)

We test whether the entry rule adds value over merely being present in high-funding names. Holding universe (151), period, per-symbol entry count, and the two-consecutive-negative exit rule fixed, we shuffle each symbol’s entry timestamps to random valid dates and re-simulate, $N = 1000$ times, seed = 42; the simulator was verified to reproduce the production engine exactly. This placebo evaluates the full two-year sample at \$200/symbol with three concurrent positions; its actual total profit of \$23.71 is therefore a full-period figure and is not directly comparable to the validation-window carry results in §4.2 (e.g. +\$47.44 on the broad universe). The actual signal’s total profit is \$23.71, versus a placebo mean of \$54.81 and median \$62.37; the actual sits at the 37.8th percentile of the placebo distribution, with a one-sided p-value of 0.622—not significant at 5% (Figure 1). In this sample the funding entry rule does not beat random entry timing. In performance-attribution language, the strategy’s return is beta to a carry factor, not alpha from timing skill, consistent with the crypto-carry factor documented by Schmeling et al. (2023)—the Henriksson and Merton (1981) timing-versus-selection question answered here non-parametrically, by randomization. We state this as an absence of evidence for skill in this sample, not as a proof that skill is impossible.

5. Limitations

We list all identified weaknesses without pruning.

Triangular track. (A1) Single exchange (Binance); cross-exchange arbitrage is out of scope. (A2) The calendar window is short—2.74 days—despite the large tick count, so seasonality and stress events are under-sampled. (A3) Only five curated, connected markets; wider triangle graphs are unexplored. (A4) The 0.2-second simultaneity threshold is a modeling choice that sets the detection/near-miss boundary; other latency assumptions would move the line. (A5) Taker fee of 0.1%; maker/VIP schedules would shift the boundary.

Funding track. (B1) Strong regime dependence: \$196.94/yr (validation) versus \$27.55/yr (full two years) for the same configuration. (B2) Survivorship bias: the universe is fixed to the current top-

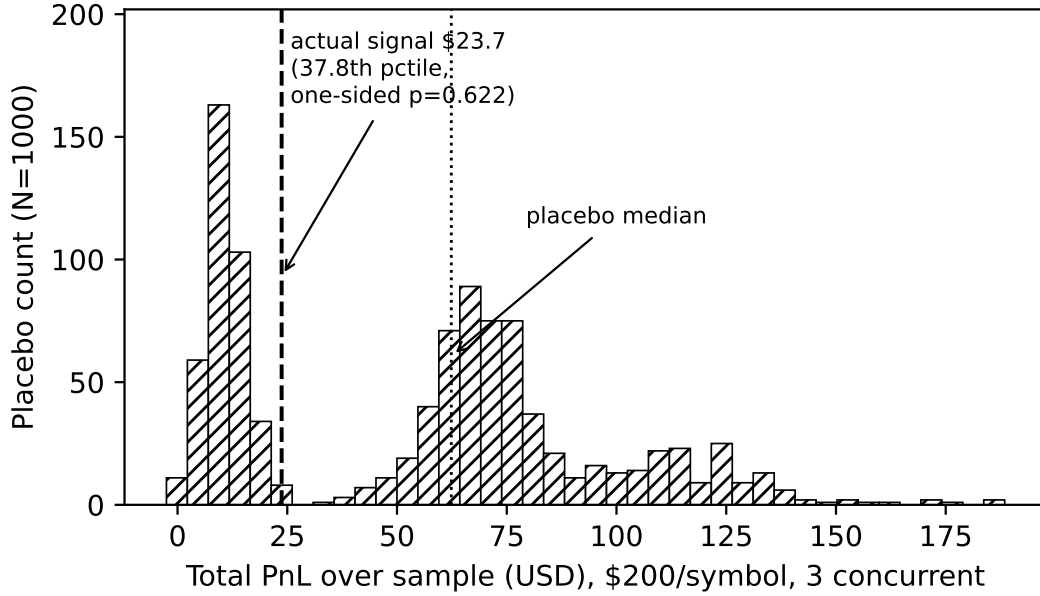


Figure 1: The funding entry signal versus random entry timing. Histogram of total PnL across $N=1000$ placebos that hold universe, period, per-symbol entry count, and exit rule fixed while randomizing entry timing (seed=42). The actual signal (dashed line) falls at the 37.8th percentile of the placebo distribution (one-sided $p=0.622$). Source: Author's calculations.

100 by volume and replayed backward, excluding delisted/pre-listing names. (B3) Idealized fills: no slippage, partial fills, funding-settlement timing, or borrow constraints; delta-neutral spot+perp execution costs are simplified; fill probability is unmeasured. (B4) Few-event dependence: removing the top three positions halves profit (to ~51–52%). (B5) Short single-cycle sample: two years, one market regime, limiting statistical power. (B6) Walk-forward combination selection is unstable across data/code revisions (e.g., the broad universe selects a 3.0-day/two-negative combination here) ; the qualitative reversal (zero→positive) is robust but the magnitude is a reference value. (B7) The reversal is a universe-selection effect (majors list vs. volume-ranked), not a count effect.

FX/CIP track. (C1) Not empirical: the FX/CIP conclusion rests on a 12-event replay fixture (engine/design verification), with no live-stream observation; we make no empirical claim from it.

Forward validation. (D1) The live paper-trading runner has opened a single funding position ; this is an in-progress forward validation, out of scope, and not a statistical result.

Methodology / reproducibility. (E1) The absence of triangular arbitrage holds under the simultaneity constraint; without it, naive measurement over-reports phantom arbitrage (Section 4.2). (E2) AI assistants were used in code, analysis, and drafting (see Acknowledgments). (E3) The placebo (Section 4.4) is a post-hoc robustness check, not pre-registered.

Placebo track. (F1) The placebo preserves per-symbol entry counts and shuffles only timing, so it

isolates timing skill but does not remove the exposure (“being in the name”) effect; the exposure return exists independently of the signal.

6. Conclusion and Future Work

In our sample, cryptocurrency arbitrage exists but is not accessible to retail as a skill-based edge. Triangular arbitrage yielded zero executable detections over 2.74 days and 66.5 million ticks under a retail-realistic simultaneity constraint ; the positive near-misses are phantom, produced by non-simultaneous quotes. Funding-rate carry does generate positive backtested returns, but those returns are regime- and concentration-dependent, contingent on universe selection, and—most importantly—not distinguishable from random entry timing.

This reframes the expectation for our ongoing forward validation (the live paper-trading campaign). One should expect returns from carry exposure, not from the entry rule’s timing skill: the rule earns beta to a funding-carry factor rather than alpha, so the forward test should be read as a measurement of realized carry net of execution frictions (slippage, fill probability, borrow), not as a validation of signal skill. Future work should (1) extend the triangular measurement across exchanges and longer calendar windows; (2) measure realized fill probability and slippage in live paper trading to convert idealized backtest returns into executable ones; and (3) test whether any funding-timing rule can beat the carry-exposure placebo out of sample.

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Appendix A: Reproducibility

Code. All analysis code is at git commit `87d6c6b` (full 40-hex hash recorded in `snapshot.json`). The engine (`arbgraph/`), strategy rules (`arbgraph/strategies/funding.py`), and backtester (`arbgraph/backtest.py`) are covered by a passing unit-test suite (187 tests).

Triangular evidence. Read-only snapshot directory `snapshot_20260718T113912Z/` under `paper/data/` (`snapshot.json`, plus copied journals): 66,476,148 ticks, 0 executable detections, +51.6424 bp best near-miss, capture 2026-07-18T11:39:12Z, runner start 2026-07-16 02:55:34 KST (UTC+9).

Funding data. Frozen caches under `data/funding_hist/{binanceusdm,bybit}`; coverage 2024-07-16 to 2026-07-16 (729.9 days), 151 symbols (100+51).

Regeneration. `paper/scripts/x2_regen.py` (reports), `x2_reversal.py` (reversal, deterministic two-run), `x3_near_miss.py` (phantom-arbitrage experiment), `x4_placebo.py` (placebo, seed = 42). Frozen outputs in `paper/data/backtests/` and `paper/data/`.

Determinism. The backtest engine uses no randomness; the walk-forward reversal was run twice with identical output. The placebo fixes seed = 42 and reports engine-faithfulness against the production engine (true).

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